

MC Prototype CAD, FEA, Analysis

Mason Shackelford

Overview Photos

Upright

- Confidential

Fold 1

- Confidential

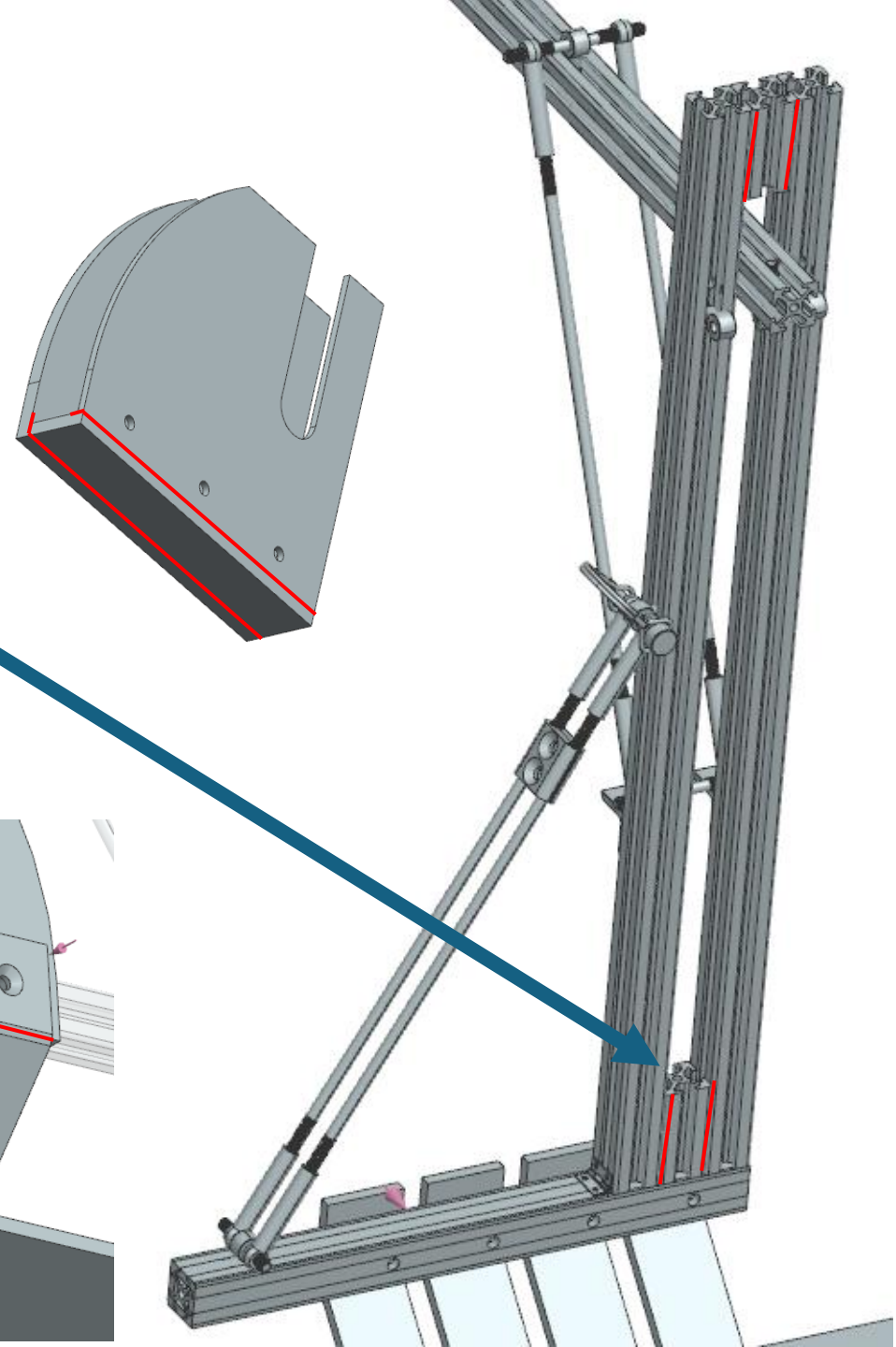
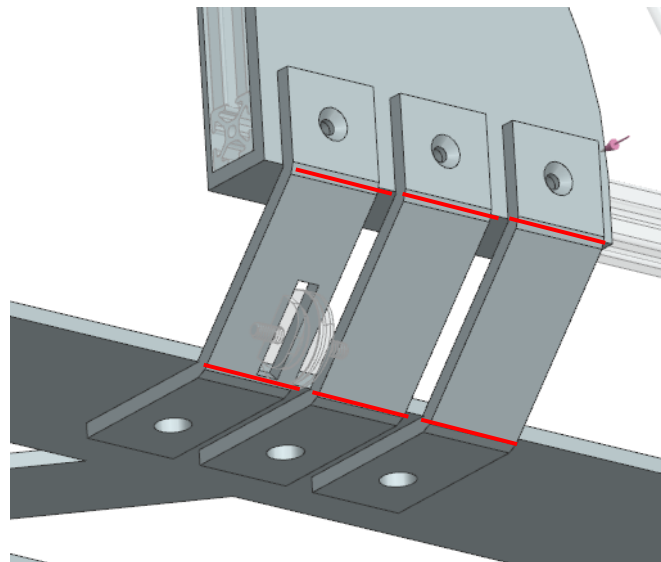
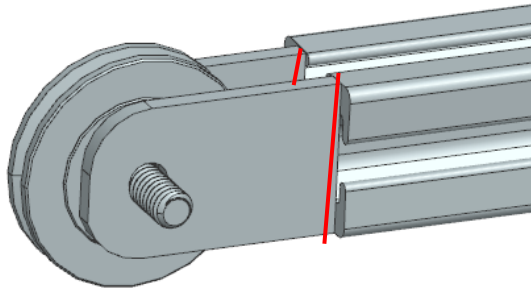
Fold 2

- Confidential

More Detail on System

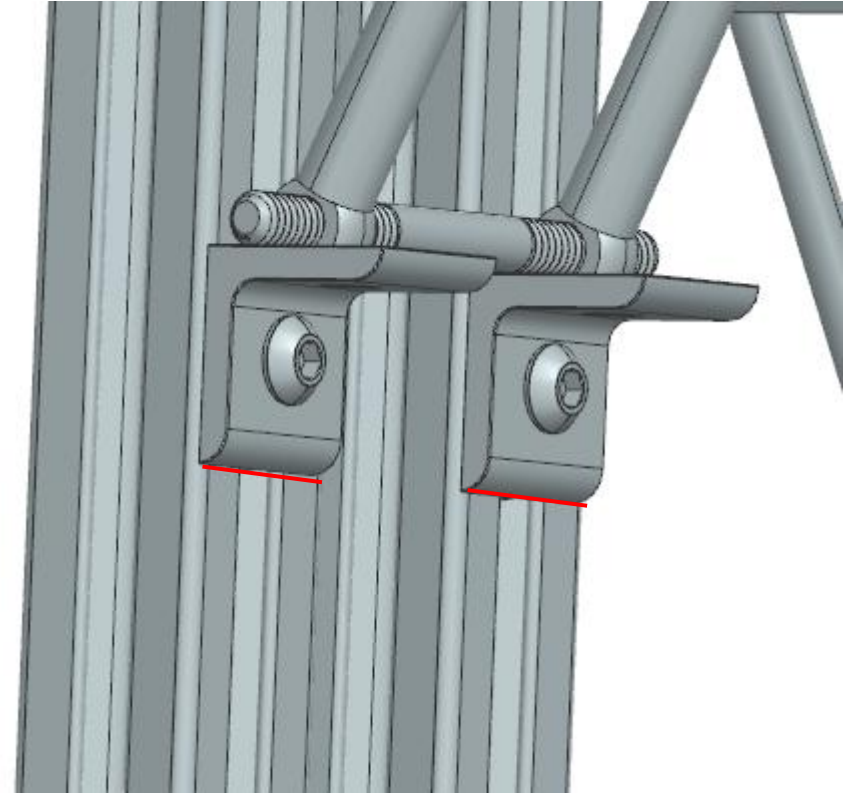
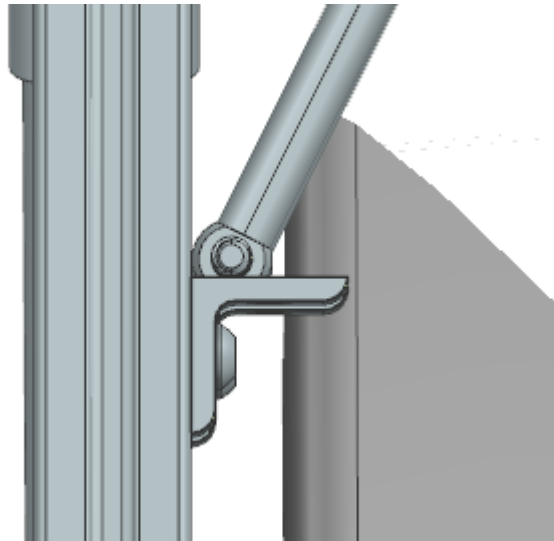
Welding

- Welds shown in red
- There should be some gap between the pieces to allow the swinging arm to fit easily between

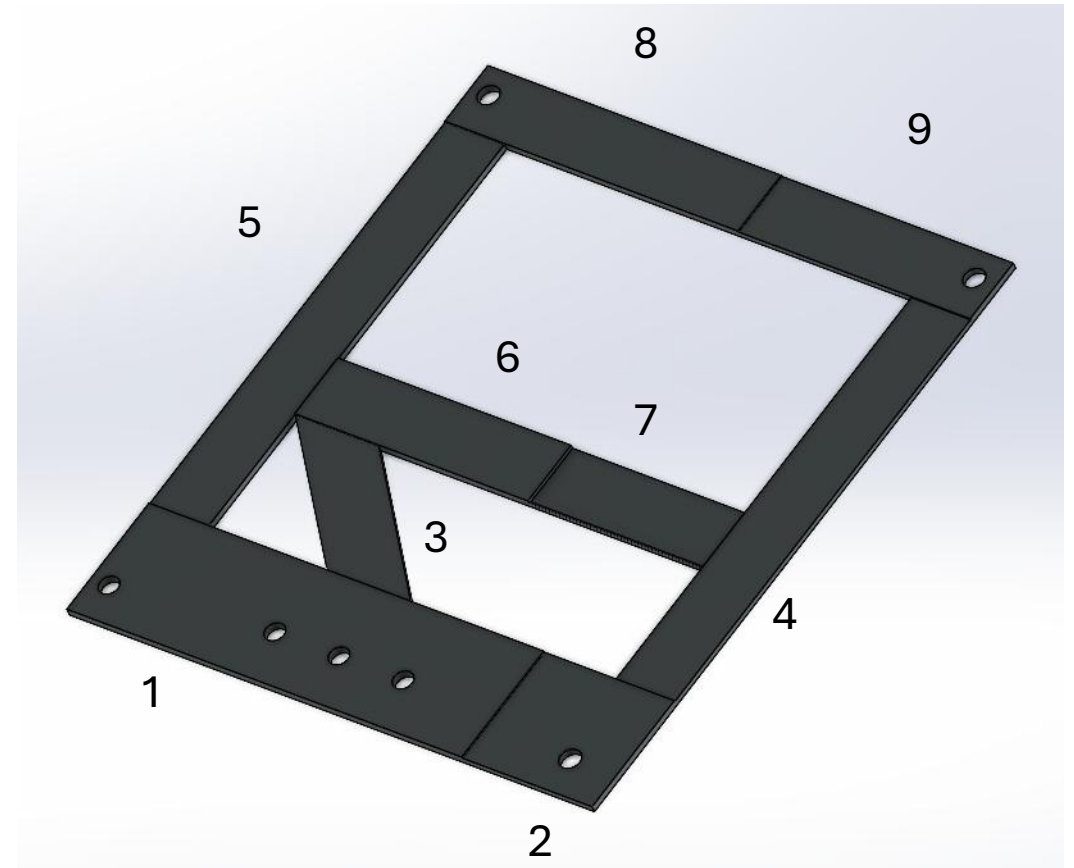
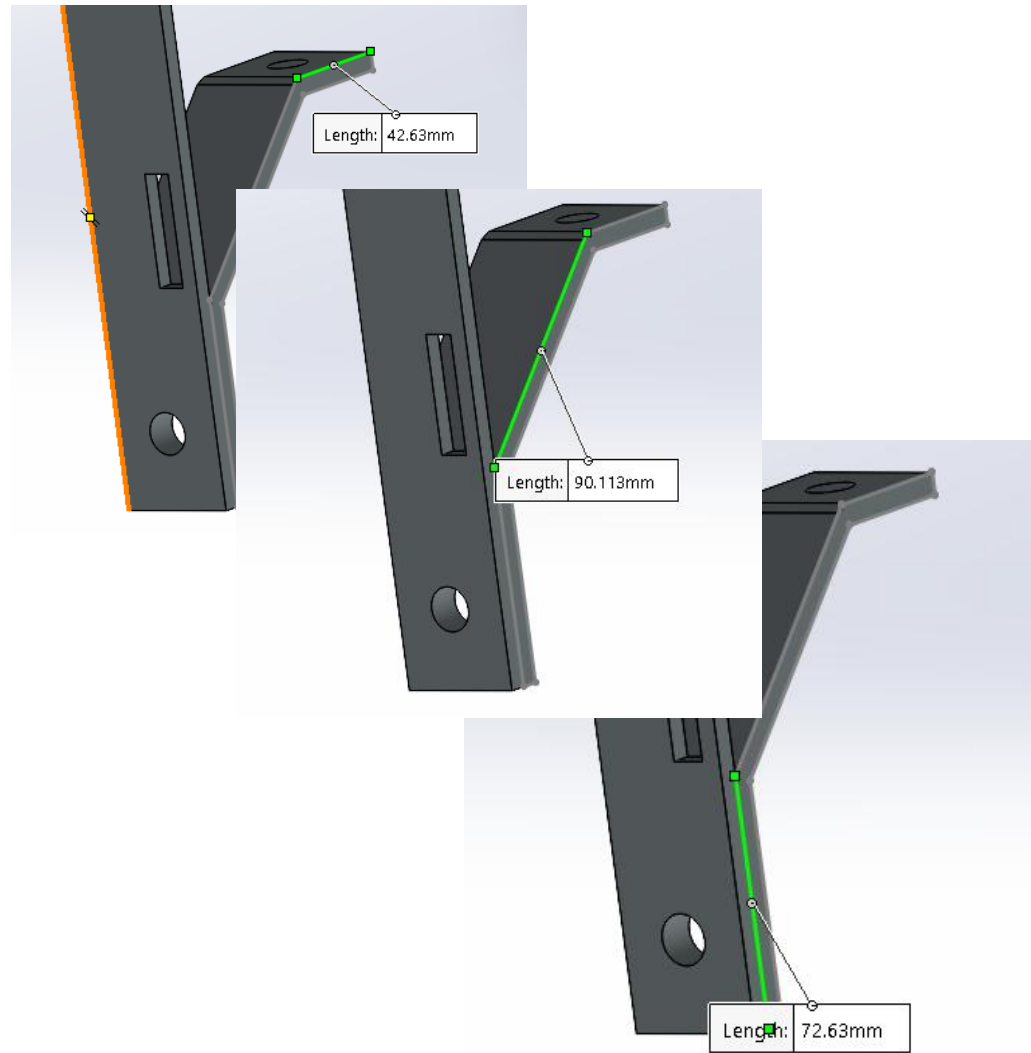


Welding

- Will probably weld these in place to prevent them from slipping
- And cut off to fit

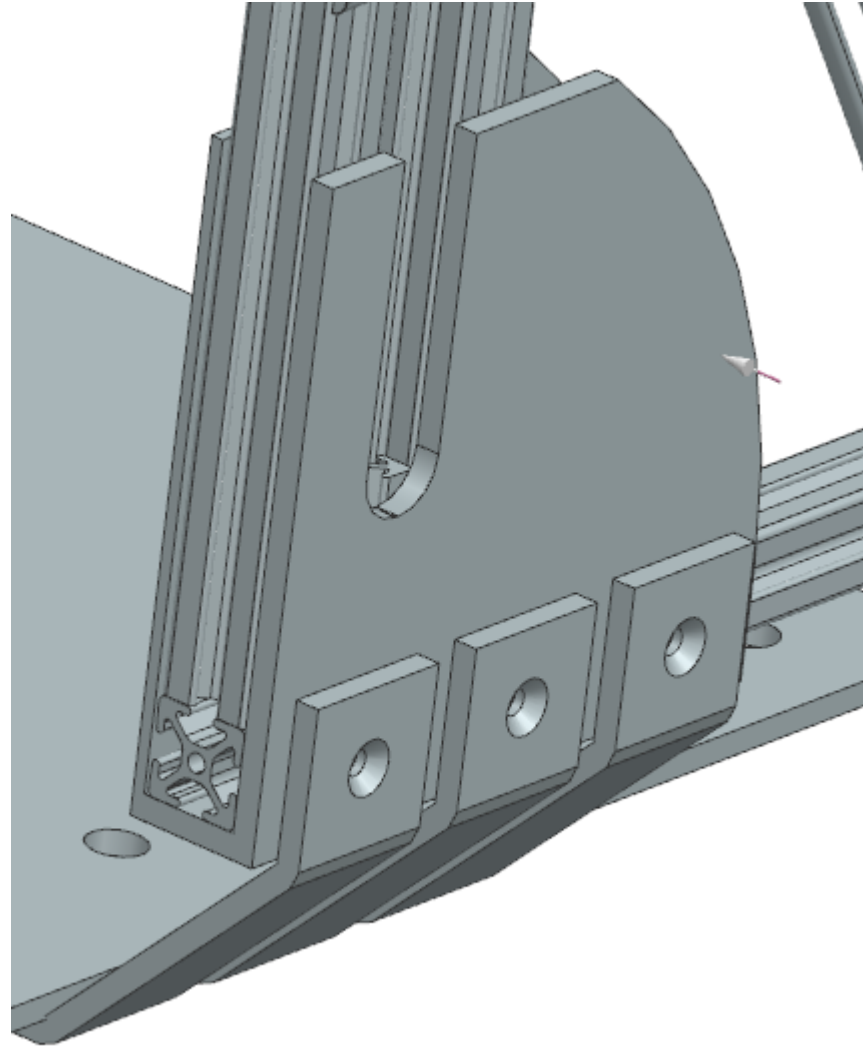


Welding



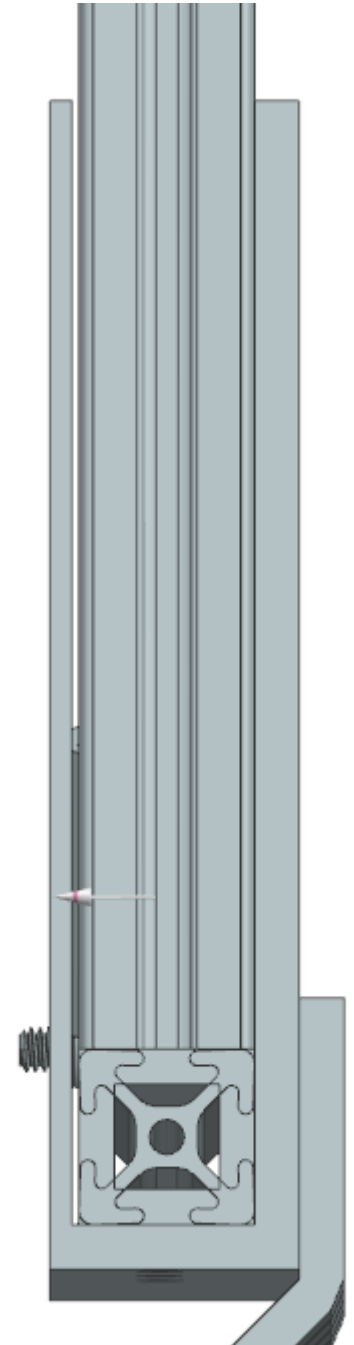
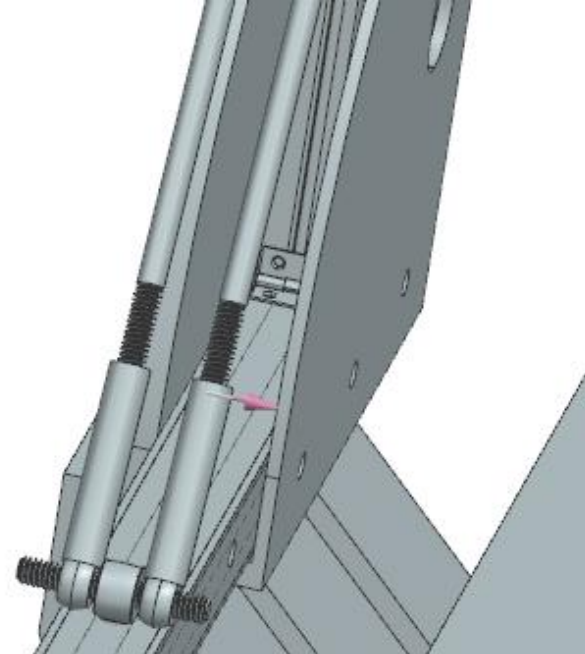
Fasteners

- These will be connected using bolts and these nuts built for t-slots:

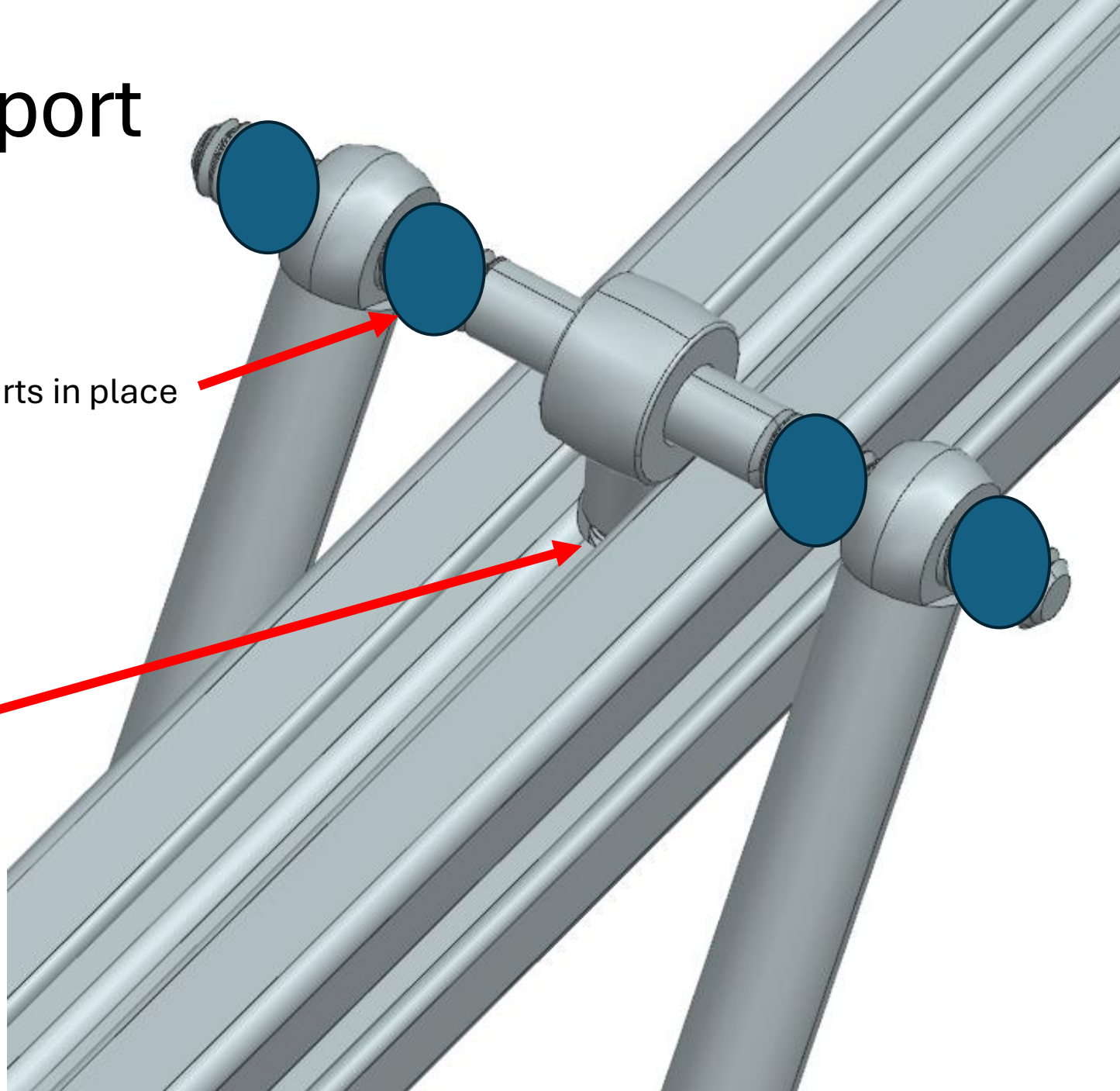


Side Support Details

- Small gap left in vertical support to allow pieces to slide up and down easily
- Hinge on vertical section is expected to deform until the vertical support contacts the sides
 - (The hinge is weak but it is by design) -> If it is too weak to support the piece pivoting on it going up and down, it can be replaced
 - To install, drill/tap size 2 (imperial) or M2 in both parts



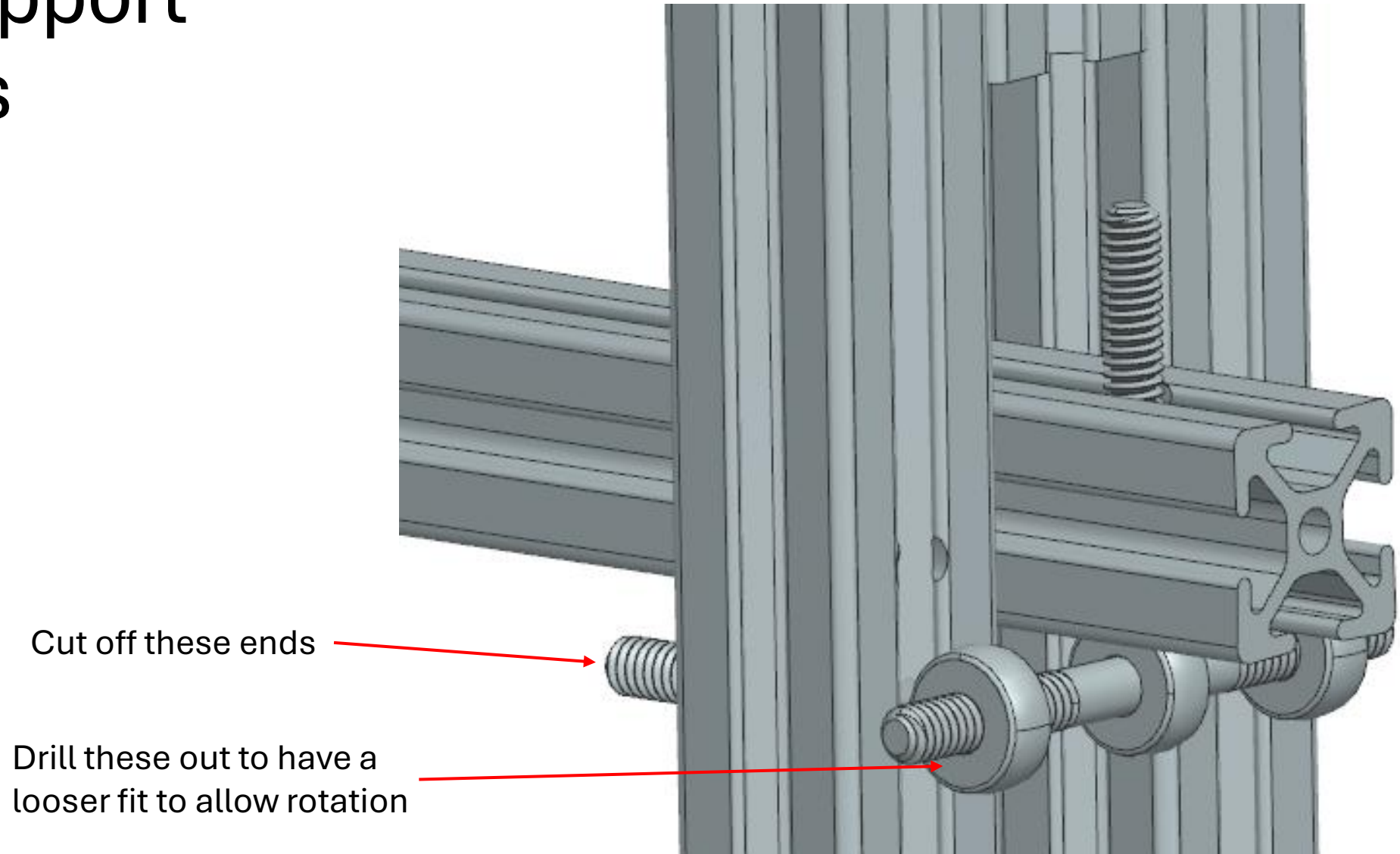
45° Support Details



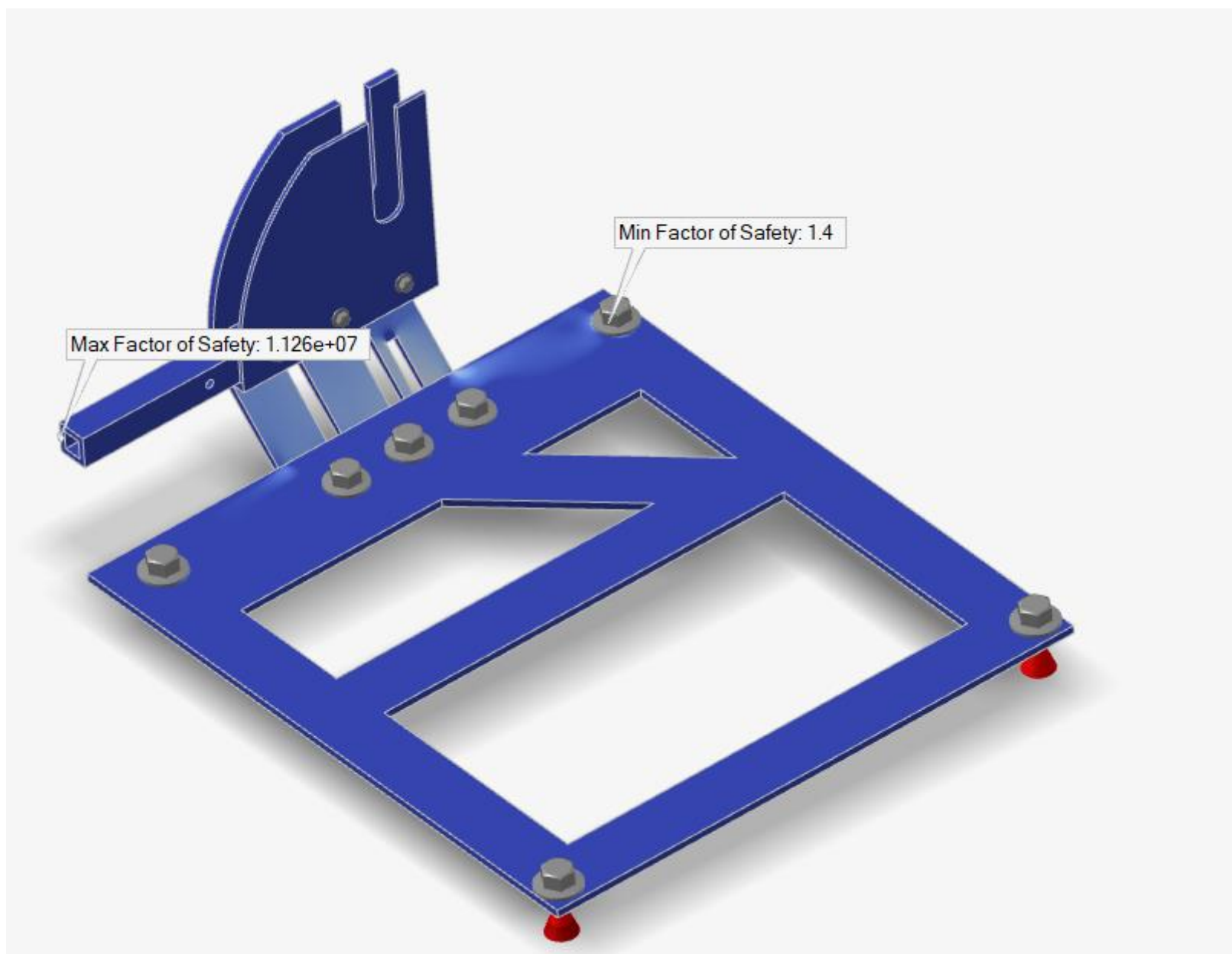
Nyloc nuts to keep parts in place

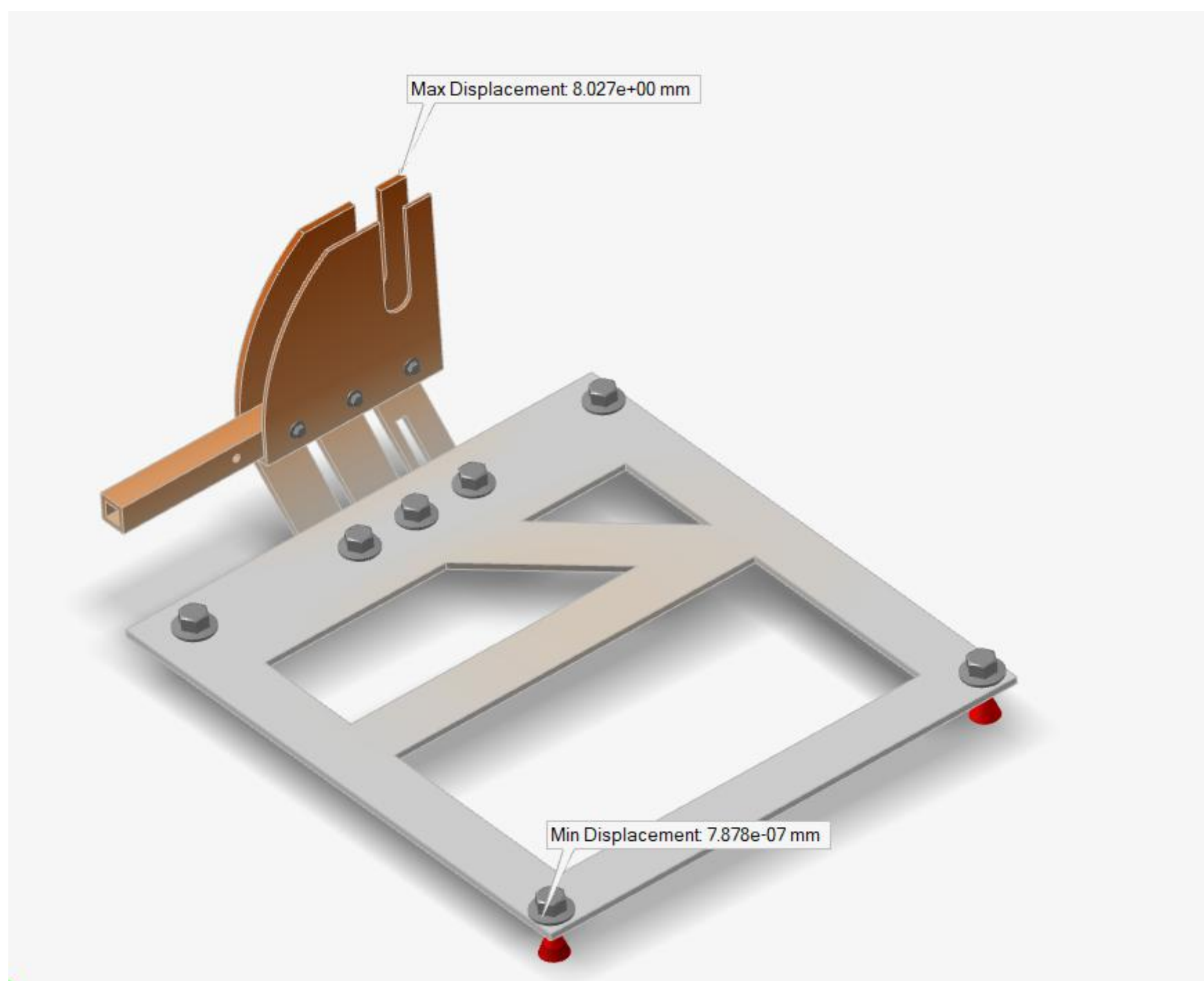
Drilled/taped hole

45° Support Details



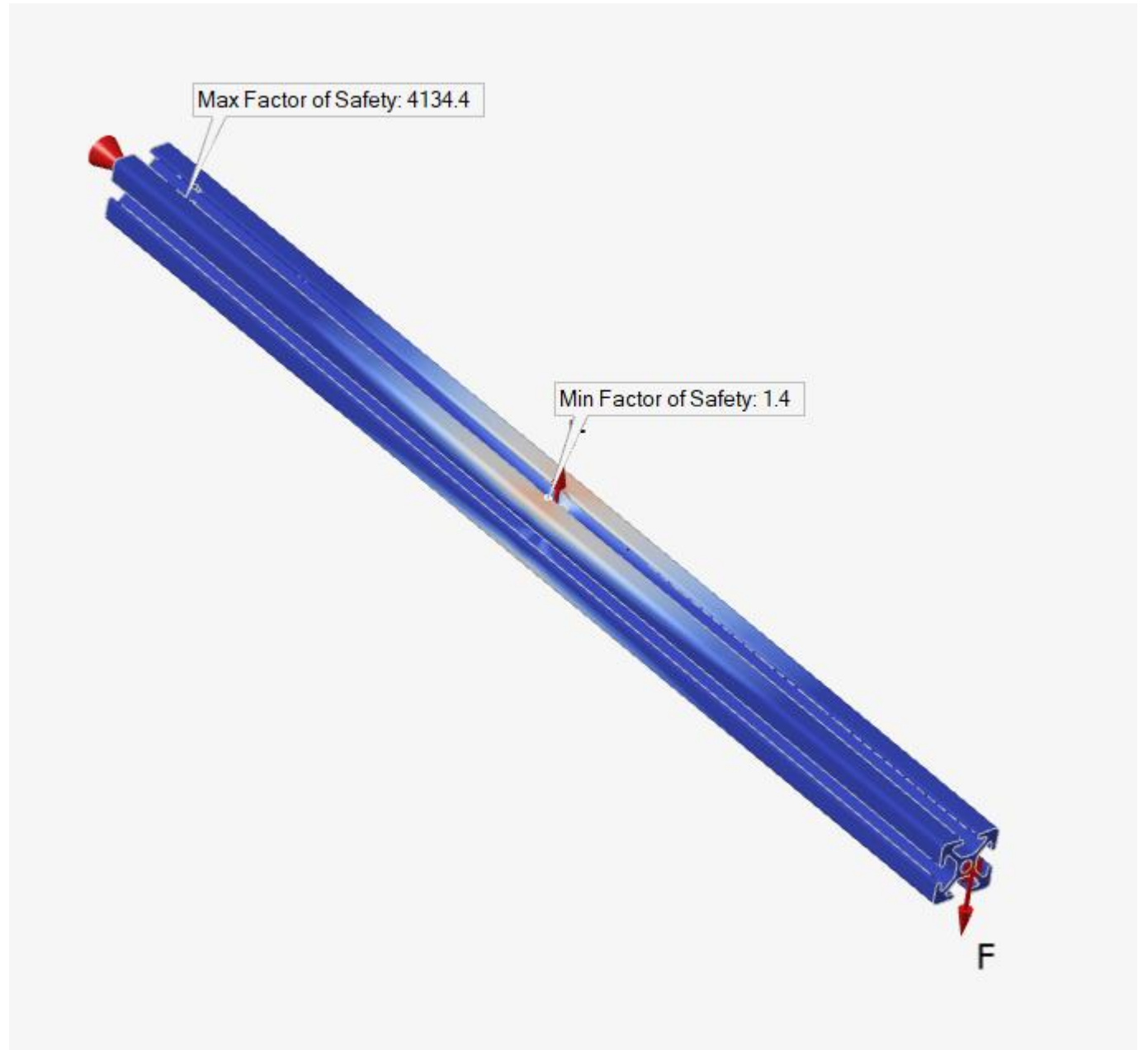
FEA





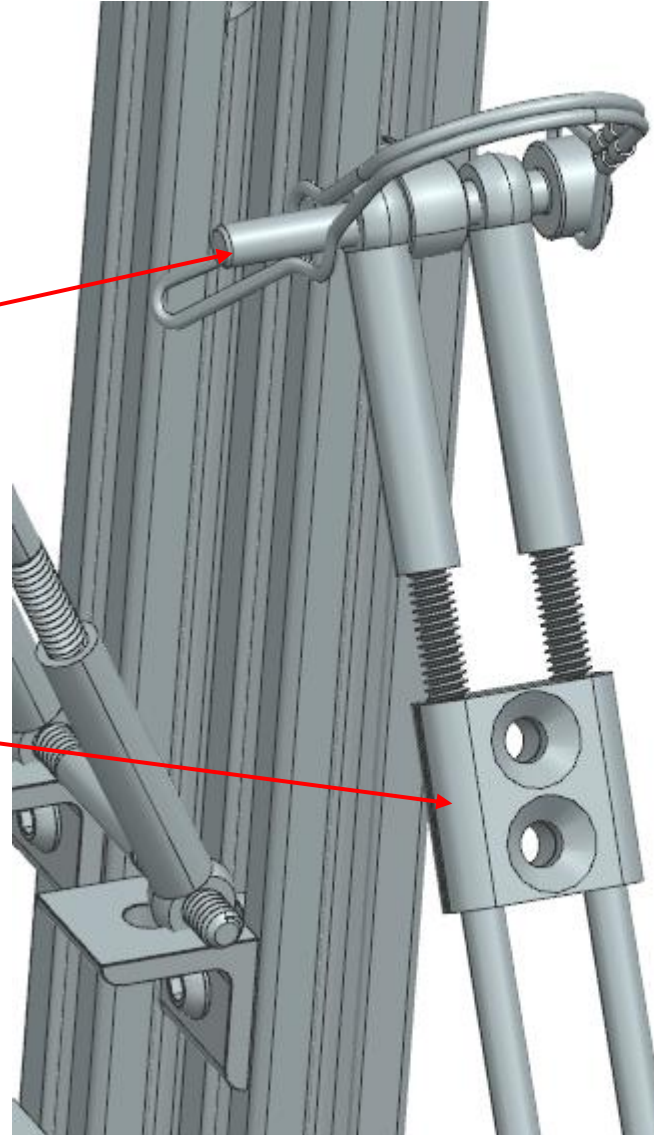
Max load on T-slot

- Middle forces
 - Up:
 - Right:
- End force:
 - Down:
- Material: 6063 Aluminum
- (yeah not exactly the right setup but conservative)



Fold 1

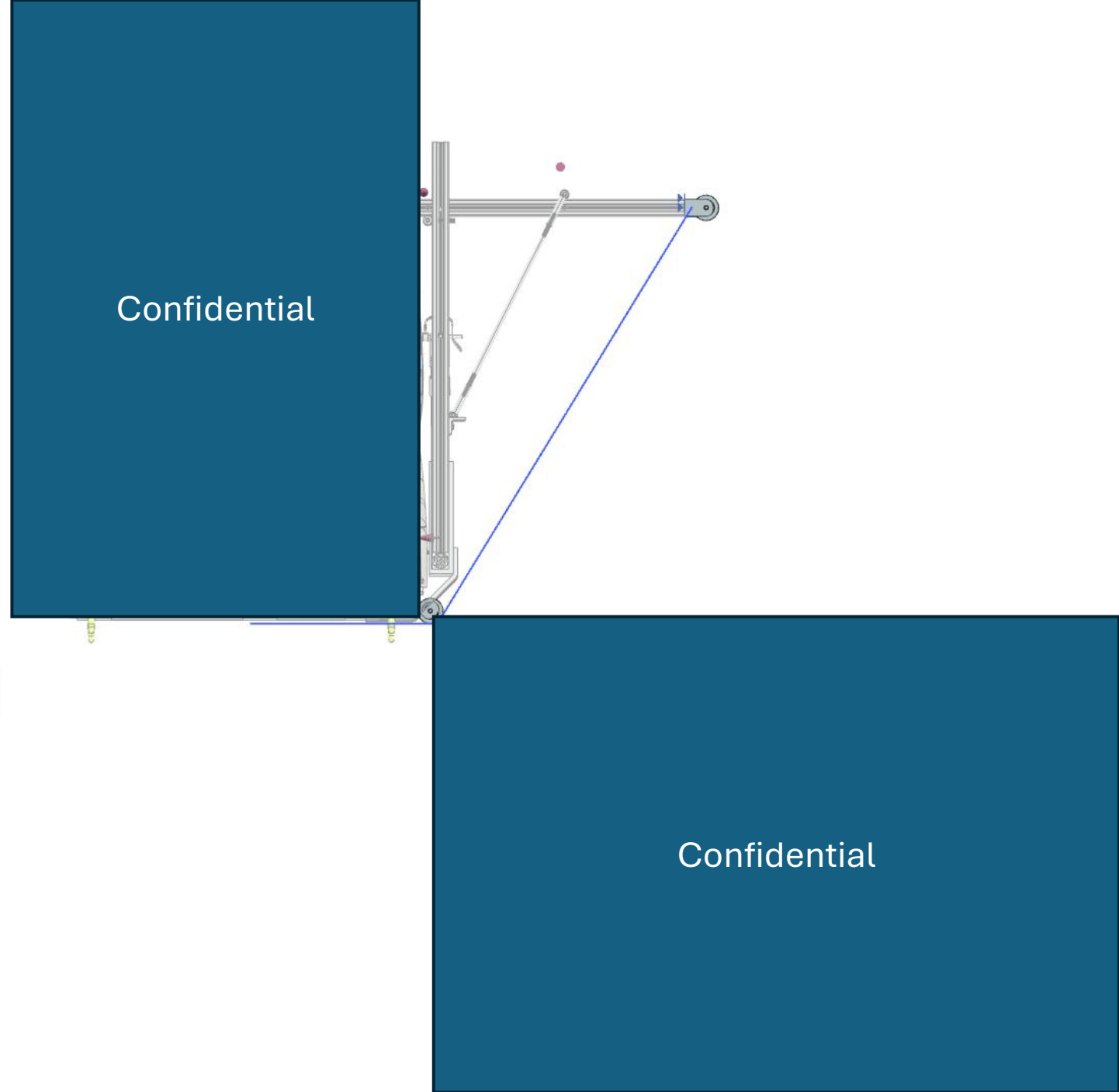
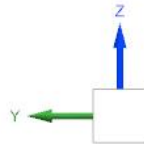
- User will fit 'key' in place to lock assembly upright
- This (3D printed) part keeps the two rods together when folded down



Pulleys

Pulleys

- Would need to cut into trim to route cable under seat to motor
 - I see no other way to route this cable
 - The motor needs to be under the seat to exist

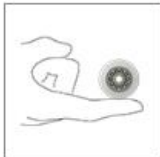


Pulleys

- Rated for 80lb (needs to handle 60lb)

Pulley for Wire Rope

Ball Bearing, 3/32" Rope Diameter, 80 lbs. Capacity, Nylon



9466T52

\$9.53 each

Quantity

1

Each

Ready for pickup in Douglasville, GA Friday by 9 am

ADD TO ORDER



3-D STEP

Download

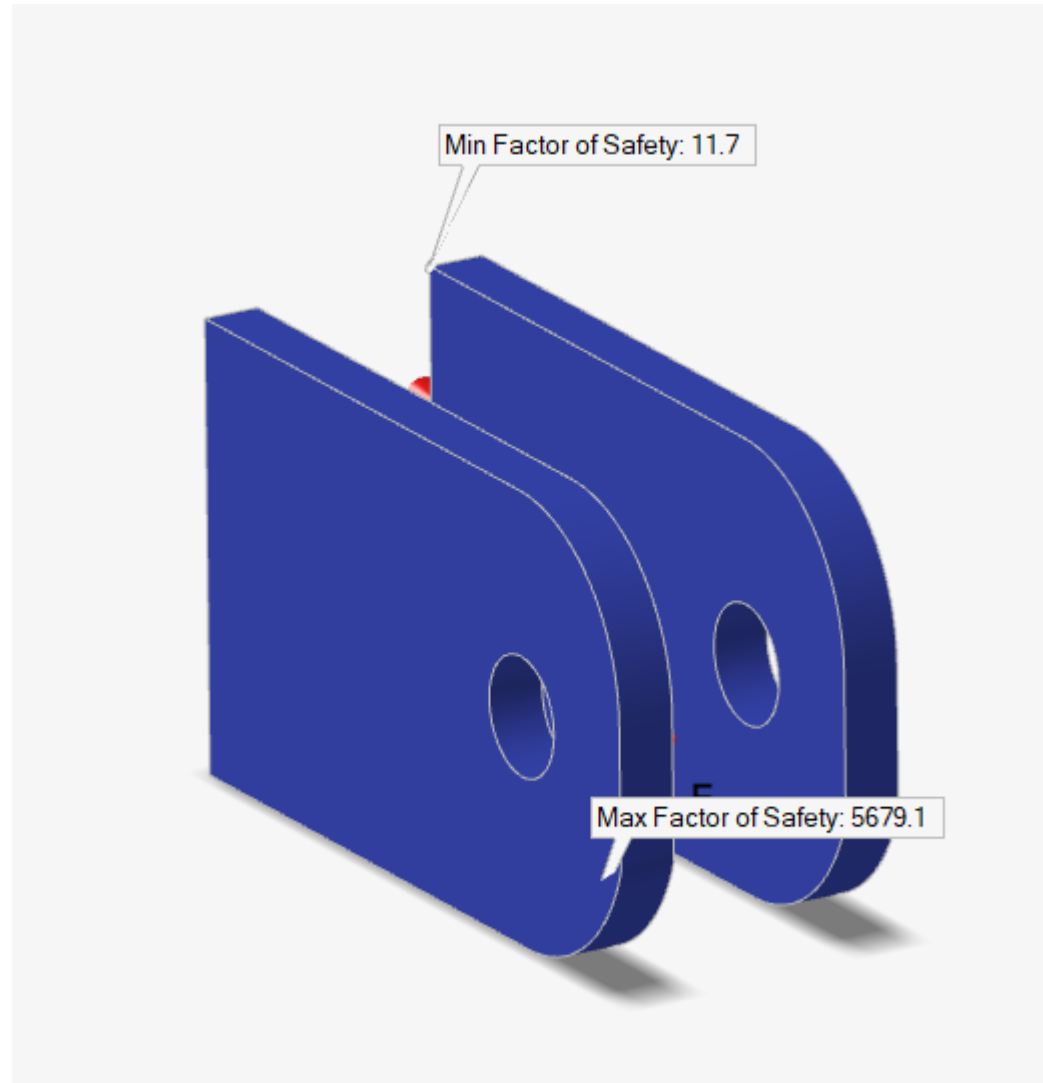
Parasolid files now available

Learn more about our [Solidworks Add-In](#)

Number of Grooves	1
Bearing Type	Ball
Style	E
For Rope Diameter	3/32"
Weight Capacity	80 lb.
OD	1 1/2"
Width	5/16"
Pulley Diameter	1 1/4"
For Shaft Diameter	1/4"

Pulley Mount

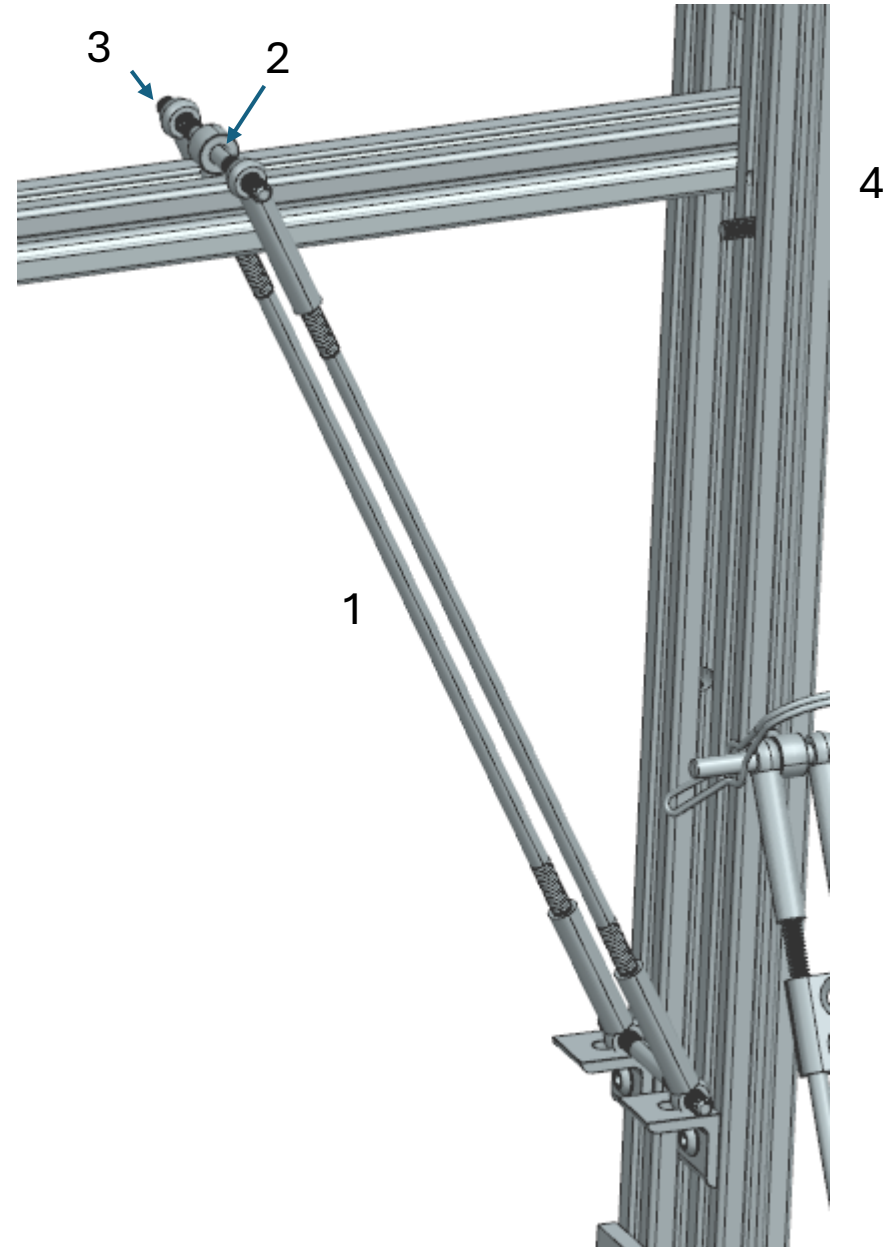
- 6061 Aluminum
- 60lb load on holes



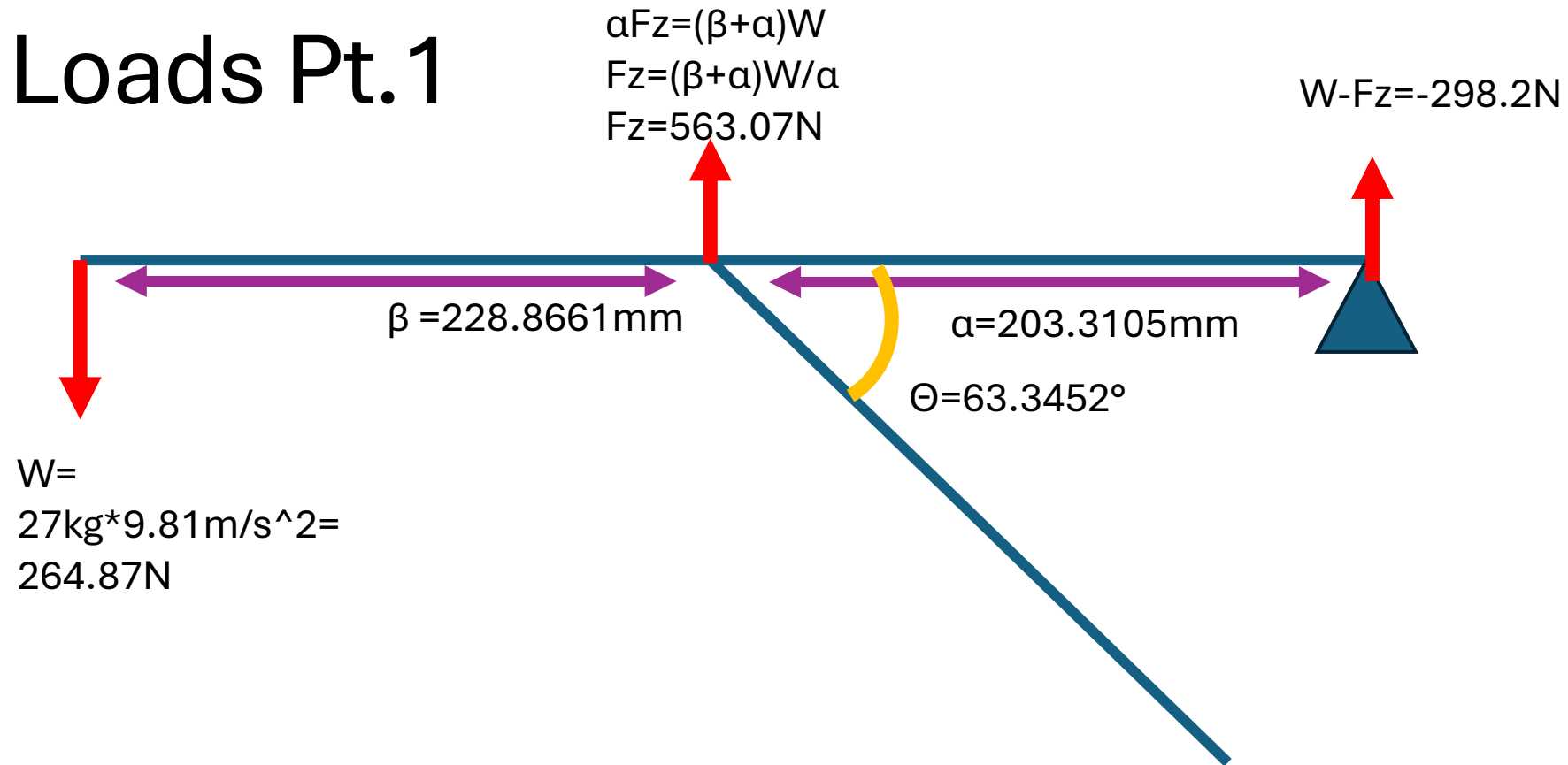
Hand Calculations

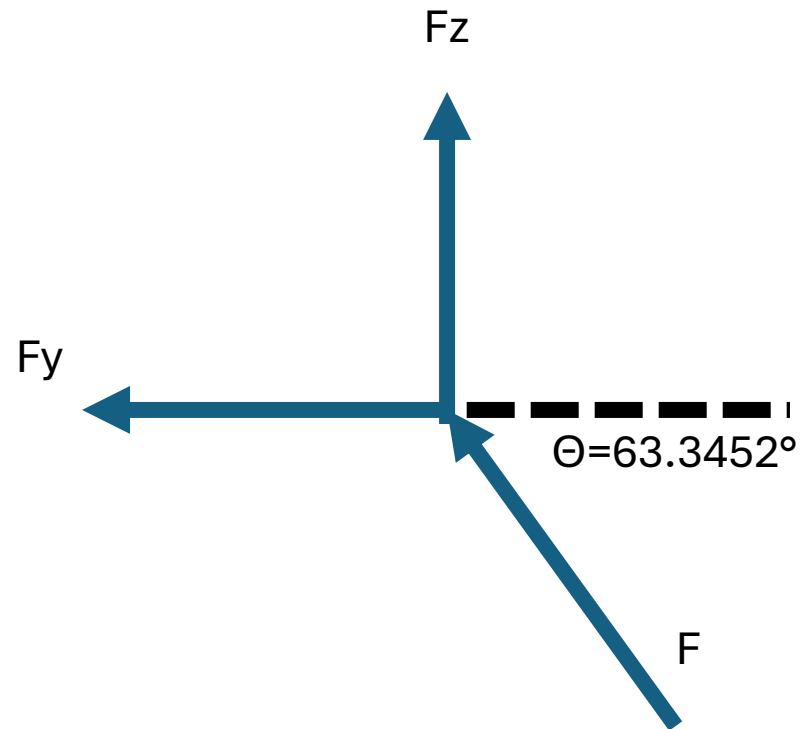
Hand Calculations

1. 45° supports (buckling)
2. Rod End Bolt (worst case)
3. Rod for 45° support
4. Vertical Support



Loads Pt.1





$$\sin\Theta = F_z/F$$
$$F = F_z/\sin\Theta = 630.03\text{N}$$

$$F_y/F = \cos\Theta$$
$$F_y = \cos\Theta F = 282.64\text{N}$$

45° Supports (Buckling)

https://www.engineeringtoolbox.com/euler-column-formula-d_1813.html

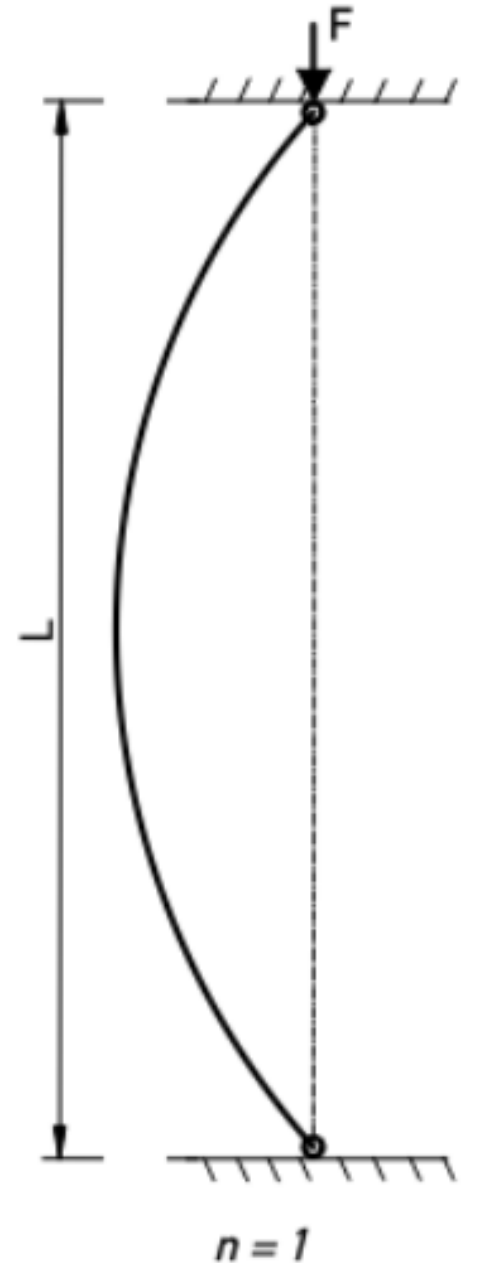
$$F_c = \frac{n\pi^2 EI}{L^2} = \frac{1\pi^2 * 69\text{GPa} * \pi(3.175\text{mm})^4}{4(304.8\text{mm})^2} * \frac{1000\text{N}}{\text{GPa mm}^2} = 585.04\text{N}$$

$$I = \frac{\pi r^4}{4}$$

$$FoS = \frac{F_c}{F/2} = \frac{585.04\text{N}}{630.03\text{N}/2} = 1.86$$

Divided by 2 because
there are 2 rods

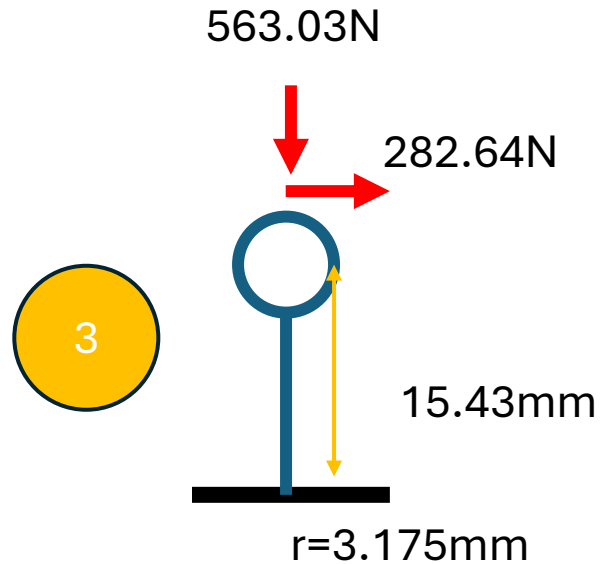
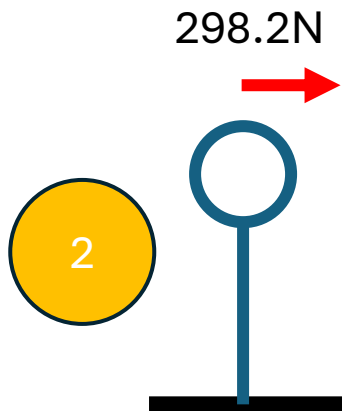
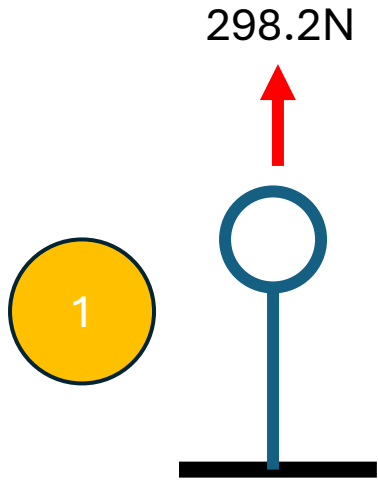
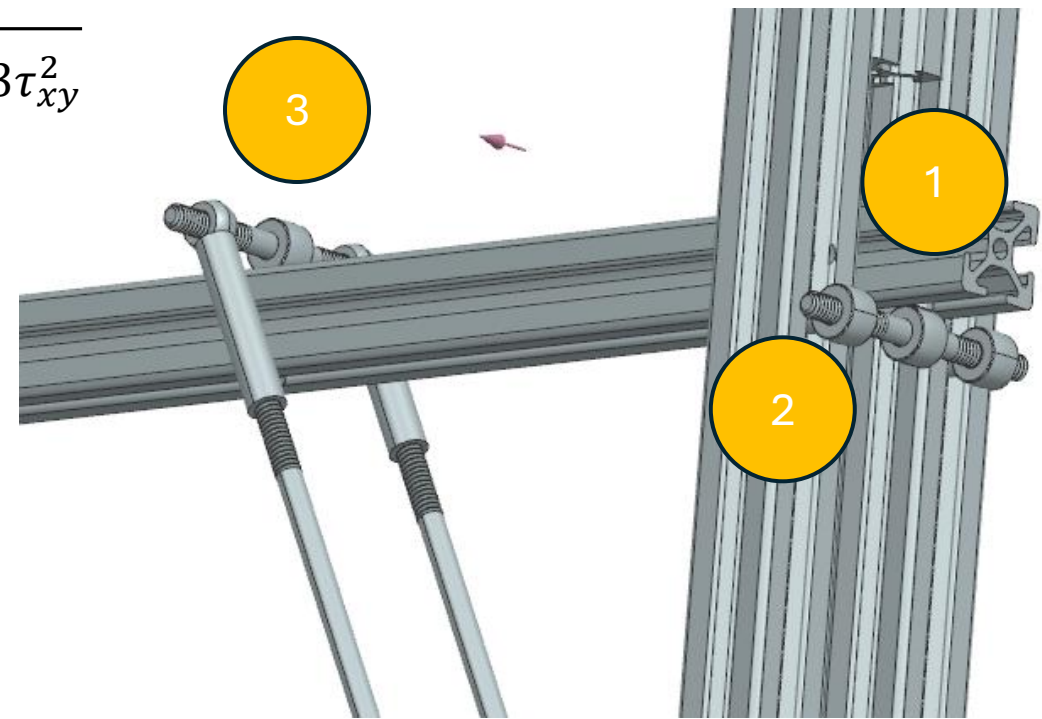
Aluminum 6061-T6; 6061-T651



Rod End Bolt

$$\sigma_{VM} = \sqrt{\sigma_{xx}^2 + \sigma_{yy}^2 - \sigma_{xx}\sigma_{yy} + 3\tau_{xy}^2}$$

Only bolt 3 here is considered since it will obviously be the first to fail



$$\sigma_x = \frac{563.03N}{\pi(3.175mm)^2} = 17.78MPa$$

$$\tau_{xy} = \frac{282.64N}{\pi(3.175mm)^2} = 8.92MPa$$

$$\sigma_{ultimate} = 861.84MPa$$

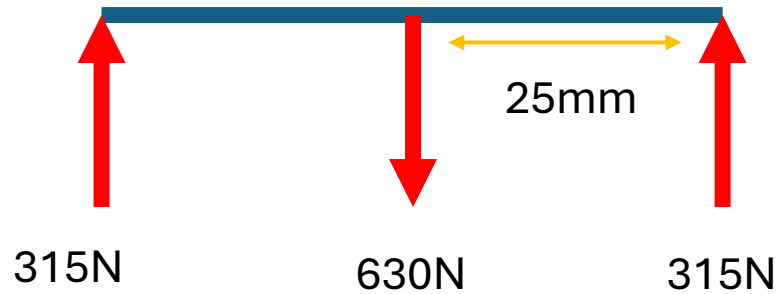
$$\sigma_{bend,x} = \frac{Mc}{I} = \frac{282.64N * 15.43mm * 3.175mm}{\frac{\pi}{4} * (3.175mm)^4} = 173.49MPa$$

Von mises: $\sigma_{VM} = \sqrt{\sigma_{xx}^2 + \sigma_{yy}^2 - \sigma_{xx}\sigma_{yy} + 3\tau_{xy}^2}$

$$\sigma' = \sqrt{(\sigma_x + \sigma_{bend,x})^2 + 3(\tau_{xy})^2} = 363.71MPa$$

$$FoS = \frac{\sigma_{yield}}{\sigma'} = 2.37$$

Rod for 45° Support



Assume it has strength greater than or equal to weakest fastener grade: Grade 8.8 at σ_{yield} 640 MPa

Description	Figure	Second moment of area	Comment
A filled circular area of radius r		$I_x = \frac{\pi}{4} r^4$ $I_y = \frac{\pi}{4} r^4 \quad [1]$ $I_z = \frac{\pi}{2} r^4$	I_z is the second polar moment of area .

$$\tau_{xy} = \frac{315N}{\pi(3.175)^2} = 9.95MPa$$

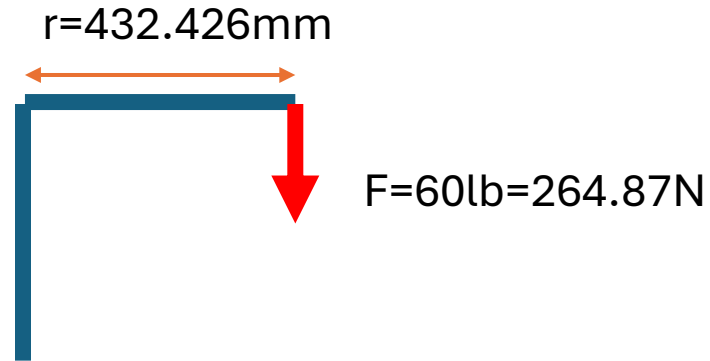
$$\sigma_b = \frac{2My}{I_c} = \frac{(2 * 315N * 25mm * 3.175mm)}{\frac{\pi}{4} (3.175mm)^4} = 626.56MPa$$

$$\sigma' = \sqrt{\sigma_b^2 + 3\tau_{xy}^2} = 627.5MPa$$

$$FoS = \frac{\sigma_{yield}}{\sigma'} = 1.02$$

Note: if this fails, it is easy to replace with a stronger option

Vertical Support



$$M = F * r = 264.87\text{N} * 432.426\text{mm}$$

$$\sigma_b = \frac{My}{I_c} = M * \frac{12.5\text{mm}}{17487.6745\text{mm}^4 * 2} = 40.93\text{MPa}$$

Second moment of area
calculated in SolidWorks

Number of beams

Width of T-slot is 25mm

$$\sigma_{yield} = 145\text{MPa}$$

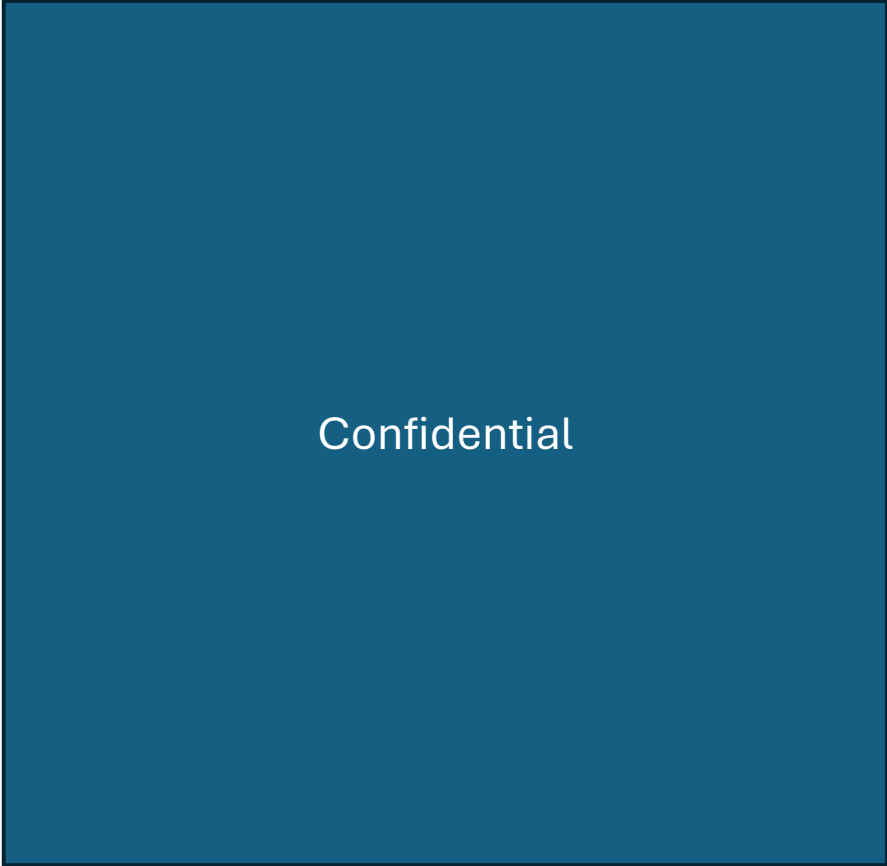
<https://www.matweb.com/search/DataSheet.aspx?MatGUID=79875d1b30c94af39029470988004fb6>

$$FoS = \frac{\sigma_{yield}}{\sigma_b} = 3.54$$

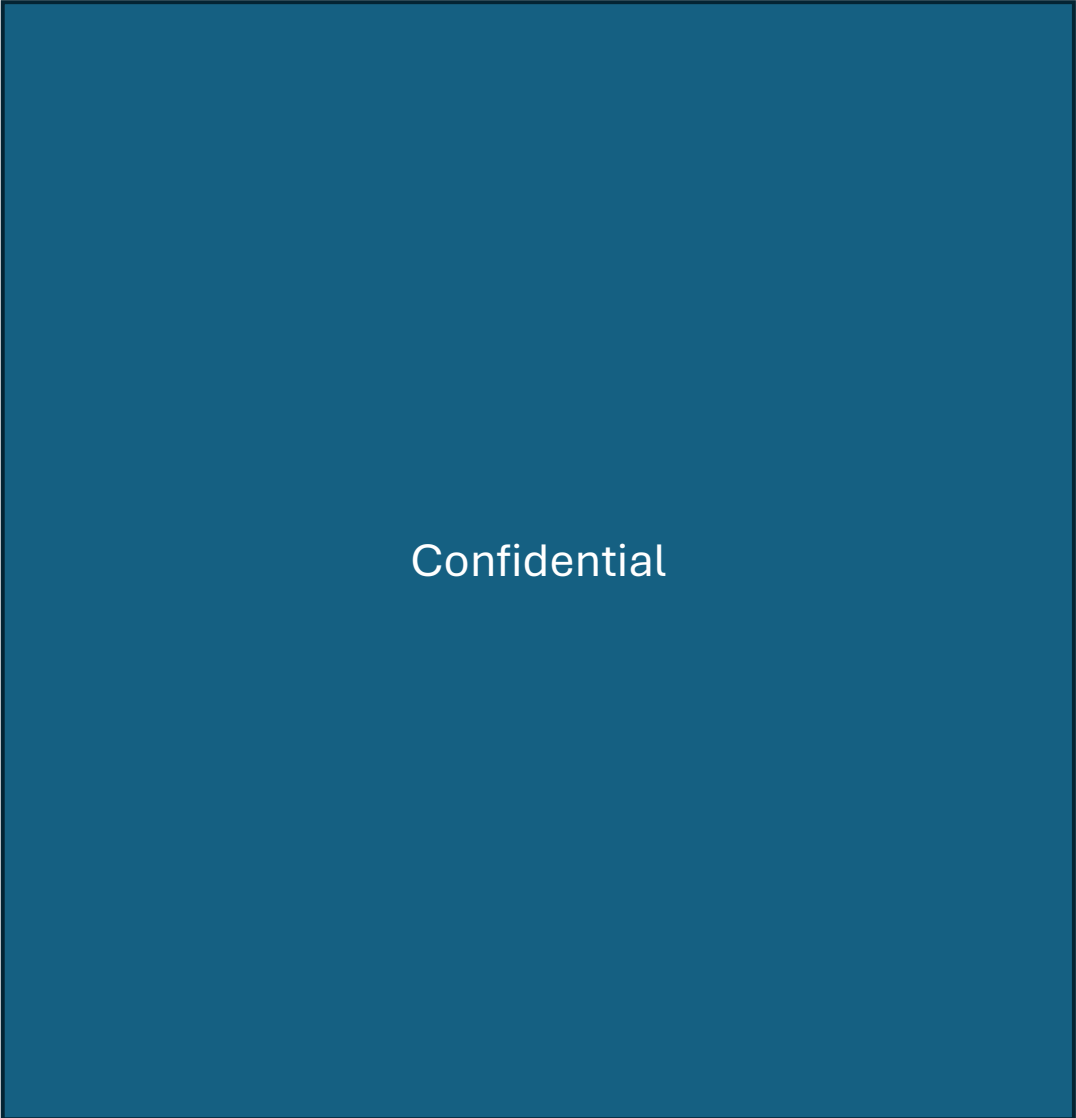
Fitment Issues

Seatbelt Part

- The seatbelt part may be an issue?



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BOM

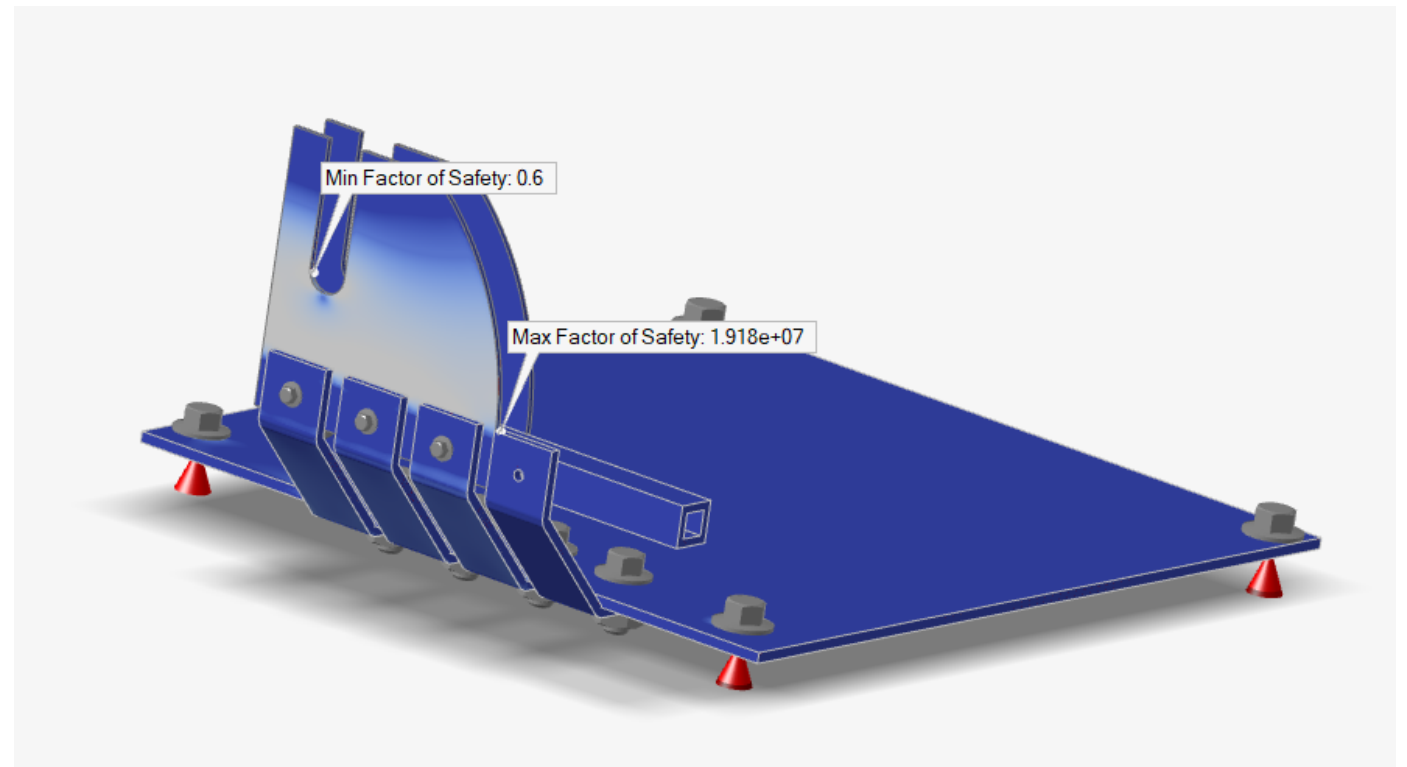
- https://gtvault.sharepoint.com/:x:/s/ECE/EcoCAR/IQDrzrVUKhcvTZduIMjvDLMzAQ_aPXZBBRcs_65Yqt2Rs8o?e=H3vpRc

Failures

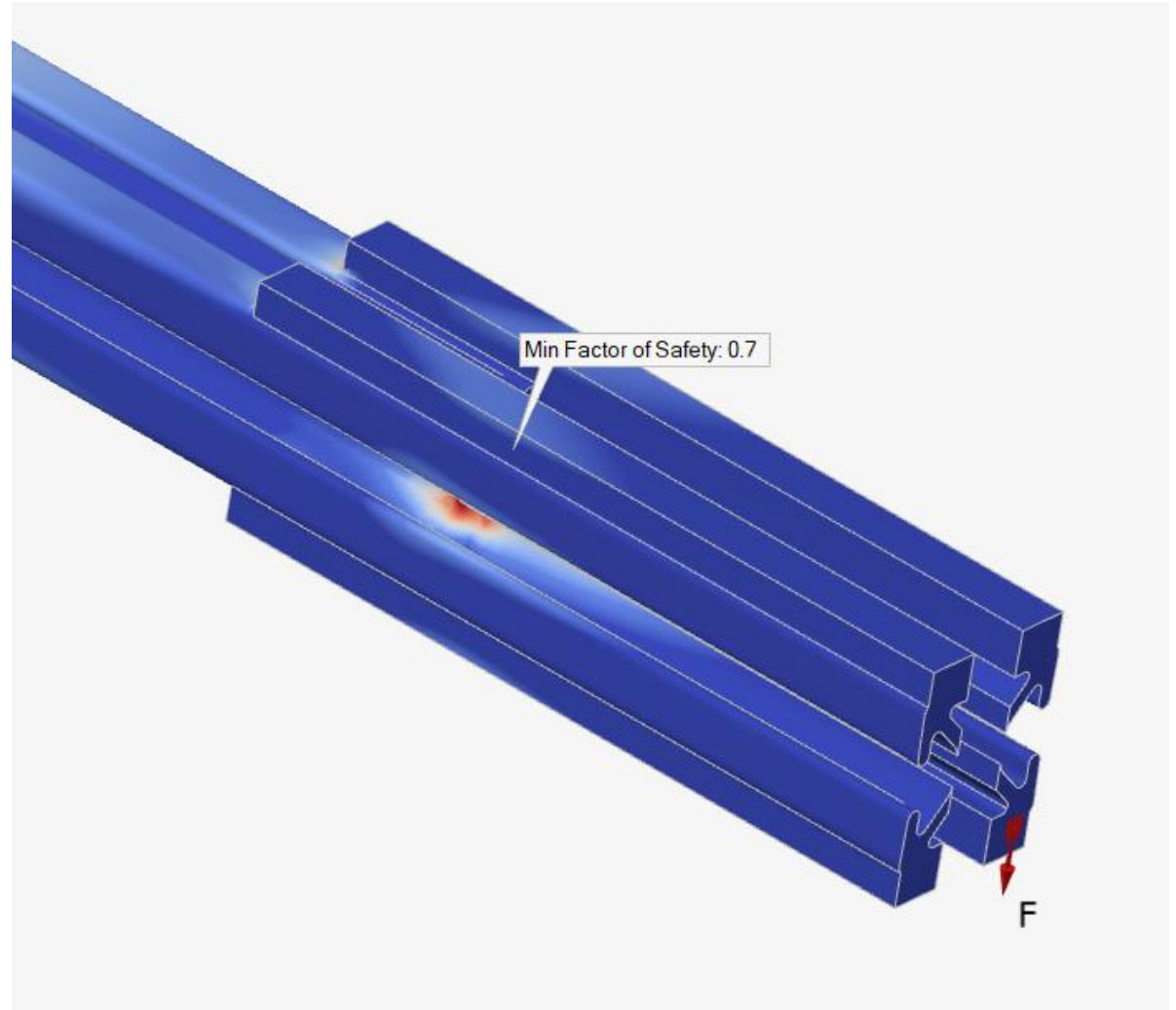
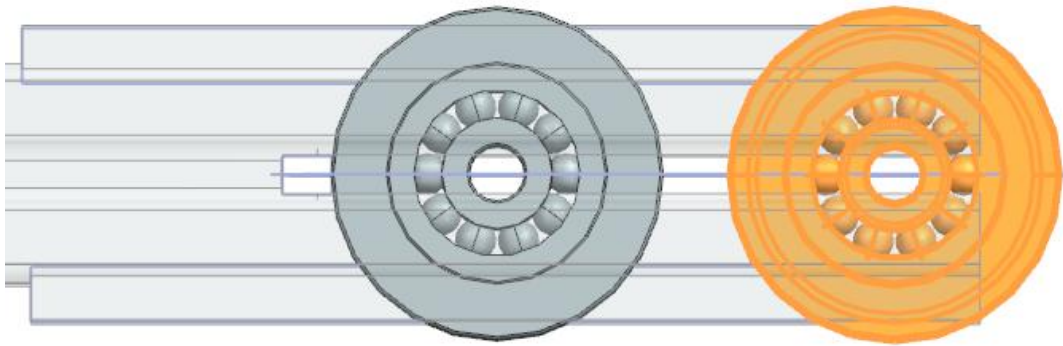
Explanations on why things are so beefy or the way they are

Why do we need 1/4" steel on side support?

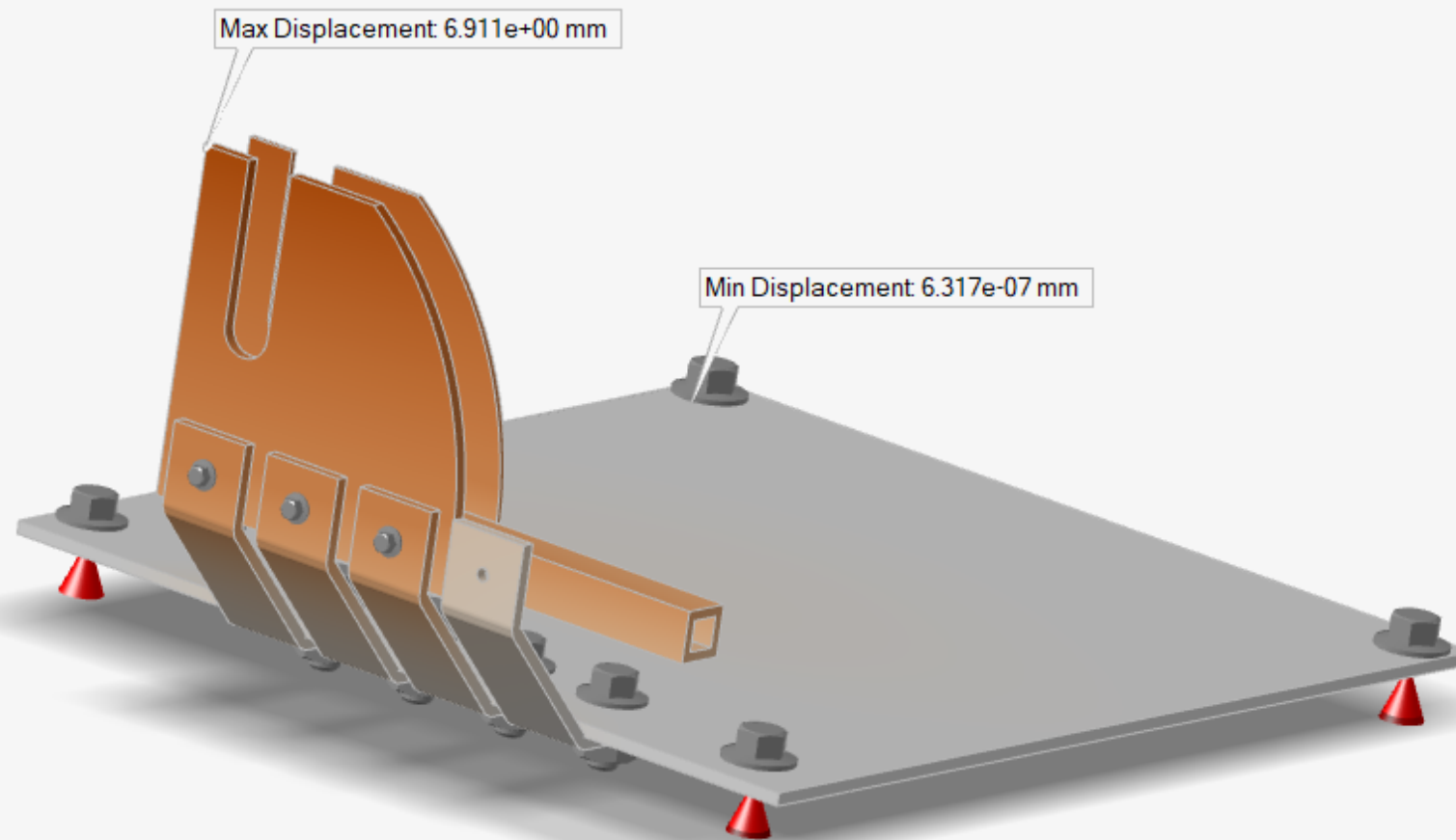
- With 1/8" steel on the outer plate -> fails
- Replaced with 1/4" steel



Can't fit pulleys inside t-slot



Is 7mm of displacement too much for a 60lb load for customer satisfaction?



Analysis Explorer

Run

Prototype_V2_2 (7)

Load Case

Load Case 1

Result Types

Displacement

Displacement: Mag

Max: 2.474e+01 mm

6.911e+00 mm

6.220e+00 mm

5.529e+00 mm

4.838e+00 mm

4.147e+00 mm

3.456e+00 mm

2.764e+00 mm

2.073e+00 mm

1.382e+00 mm

6.911e-01 mm

6.317e-07 mm

Min: 6.317e-07 mm

Show

Callouts

Min/Max

Compare Results

